

Internal calibration (IC)	The term is applied to diverse procedures. The calibration is achieved with a calibrant under different forms in the working samples. Conversely, one calibration function is obtained for each working sample to be quantified. Recently novel procedures have been developed for MS-based analysis and are detailed in Section 1.5.
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As briefly mentioned before, the analyte nature, the availability of the working sample material and the calibration material influence the selected type of calibration. This can be summarized by this simple table leading to at least four different basic configurations.

		Matrix	
		Authentic	Surrogate
Analyte	Authentic	Yes	Yes
	Surrogate	Yes	Yes

Table 1.1 attempts to classify different calibration modes, external *versus* internal, commonly used in the laboratory, including the advantages (pros) and limitations (cons) for each. As illustrated, external calibration (EC) methodologies depend on the availability of both analyte and matrix. For the procedure called in-sample calibration (ISC) there is no need to select a particular calibration matrix as the working sample matrix is used. Still remains the question of the analyte’s availability. The abbreviation ISC is introduced to make the difference with internal calibration.

1.3 External Calibration (EC)

1.3.1 Authentic Analyte in Authentic Matrix: MMEC

External calibration (EC) corresponds to the most often-used operating procedure because it allows the rational determination of several routine samples with one pre-determined calibration function $Y = f(X)$. The first situation, sometimes called matrix-matched external calibration (MMEC), represents a good metrological quantification approach and is extensively discussed in the major international guidelines to validate bioanalytical methods [6].

With exogenous substances, such as rare pollutant chemicals, a blank matrix is generally available and permits EC with authentic analyte in a representative matrix. On the other hand, with endogenous compounds at endogenous concentration, such as vitamins in foods, other approaches should be explored to overcome the absence of an analyte-free matrix. In this complicated context, alternative procedures have been proposed, such as background subtraction or the use of surrogate matrices and/or analytes as described below.

Table 1.1 Proposals for a classification of calibration procedures.

External calibration (EC)				
Ref.	Authentic analyte		Surrogate standard ^{a)}	
Matrix	Authentic	Surrogate	Authentic	Surrogate
Method	Matrix-matched (MMEC) ^{b)}	Surrogate matrix	Surrogate analyte	Surrogate analyte and matrix
Pros	Matrix effect and selectivity close to sample.	Suitable for low concentration compounds.	LOQ Lower than the background subtraction.	When authentic analyte difficult to obtain.
Cons	LOQ define by endogenous concentration.	Production of analyte free matrix. Possible differences in extraction recovery and matrix effect.	Accuracy depends on surrogate specificity. Additional experiment for linearity and LOQ.	Accuracy depends on surrogate specificity. High differences for recovery yield to be expected.
In-sample calibration (ISC)				
Ref.	Authentic analyte		Surrogate standard (calibrant)	
Matrix	Authentic	Partially labelled isotope analogue	Fully labelled isotope or structural analogue	
Method	Standard addition method (SAM)	Authentic	Authentic	
		Isotopic pattern deconvolution (IPD)	Internal calibration (IC)	
Pros	Same matrix effect and selectivity as the sample.	High potential for accuracy	High potential for accuracy (SIL)	
		Relying on isotopic distribution alteration.	Reduced numbers of calibrators.	
Cons	Need for large initial specimen volume.	Depends on analogue concentration and stability.	Depends on analogue concentration and stability.	
	Not easy implemented for high throughput.	Additional experiment for linearity and LOQ.	Structural analogues cannot compensate for differences in ionization.	
			Additional experiment for linearity and LOQ.	

a) Isotope labelled or structural analogue.

b) With or without background subtraction.

The use of authentic matrix for multipoint EC provides an extraction recovery yield that is close to the specimen and is commonly performed to quantify exogenous substances when a large amount of the matrix is available. In the presence of endogenous compounds, a representative pooled matrix fortified with authentic calibration standards can be prepared to estimate and remove the endogenous background signal. This approach, known as background subtraction, uses the pooled matrix-matched EC to interpolate the concentration in the working samples.

As described in Section 2.2, Z is the inverse-predicted concentration. It is obtained by inverting the equation of the calibration curve. Equation (2.24) illustrates the rationale in the case of a linear calibration curve where the slope a_1 and intercept a_0 refer to the regression parameters of the added authentic standards in the pooled authentic matrix.

However, the upper limit of quantification (ULOQ) as defined by several regulatory documents may be impaired by the blank response a_0 , because detector saturation may occur. Similarly, endogenous metabolite concentrations may vary due to intra- and inter-sample variation, leading to highly variable results when a pooled matrix is used. To overcome these drawbacks, several calibration curves using different representative pooled matrices can be prepared to select the calibration model that best covers the concentration to be analyzed. MMEC cannot always correct the matrix effect when it differs between working samples, emphasizing the importance of using an IS to correct this bias.

1.3.2 Authentic Analyte in Surrogate Matrix

As stated, a surrogate matrix could be used as a substitute to prepare calibrants with the authentic analyte or a mixture of analytes. It can be of various complexity. For instance, in bioanalysis, several matrices are proposed as surrogates, namely neat solutions, synthetic or stripped matrices.

- Neat solutions: it can be the mobile-phase solvent mixture, extraction reagents or pure water.
- Synthetic matrices: they are composed of salt, sugar and simulate authentic matrix properties, such as analyte solubility, extraction recovery and matrix effect. When the working sample matrix is comparable to water, saliva, urine, tears and cerebrospinal fluid, neat and artificial solutions can be used as surrogate matrix.
- Stripped matrices: they can be in-house made or commercially available, such as depleted human or bovine serum. Charcoal stripping removes nonpolar material such as lipid-related materials, mainly hormones and cytokines, leading to an analyte-free matrix that can be used as a blank for the preparation of calibrators. It is important to emphasize that charcoal depletion is nonselective and may result in approximate matrix similarity.

Whatever the chosen solution, it must be shown it has the same, or comparable, extraction properties as the authentic matrix.

Hence, surrogate matrices may not perfectly simulate the original matrix. To correct those matrix biases, a proper evaluation should be performed as recommended

by both FDA and EMA guidelines. To assess the applicability of any surrogate matrix the classic requirement is to compare the slopes of the calibration curves calculated with the surrogate matrix and authentic matrix. Diverse statistical treatments are available, such as analysis of variance.

But only EMA specifies how to assess the matrix similarity by using the concept of acceptance. This consists in ratioing the slope between authentic analyte in authentic matrix *versus* authentic analyte in surrogate matrix. The obtained value should be within $\pm 15\%$ of the nominal value. Example of possible procedures is fully described in Section 2.4.3 and illustrated in the worksheet named Resource D. The standard addition method (SAM) is one other dedicated tool to achieve this goal.

1.3.3 Surrogate Calibrant in Authentic Matrix

In situations where calibration is performed using a surrogate standard, it is assumed that the physicochemical properties of both authentic analyte and the surrogate calibrant are equivalent. For instance, the extraction recovery, the chromatographic retention behavior, and the instrument response should be either identical or have acceptable differences to be fully exploited. The choice of surrogate calibrant is essential to accurately quantify the authentic analyte.

For example, ICH guidelines [7] suggest using SIL molecule as surrogate calibrant in authentic matrix, while FDA guidelines [3] do not endorse this methodology. Because the calibration reference compound does not correspond to the authentic analyte, the ratio of responses between surrogate and analyte should be investigated over the desired dynamic range. Before routinely using the surrogate calibrant, the response factor RF must be evaluated as an analyte-to-calibrant ratio where X_{AA} and X_{SS} are the concentrations of authentic analyte and surrogate standard, respectively, and corresponding instrument responses:

Response factor (analyte *versus* surrogate)

$$RF = \frac{Y_{AA}}{X_{AA}} \times \frac{X_{SS}}{Y_{SS}} \quad (1.2)$$

To achieve the appropriate RF estimation, different proportions of analyte/surrogate must be investigated. For MS methods, this step is compulsory to evaluate the ionization efficiency whereas the RF must be constant over the method working domain. Another way to investigate the RF is to check if both lines are parallel. It consists in comparing the slopes of the authentic analyte line and the surrogate, both performed in the same pooled matrix.

Additionally, if the RF is not constant over the validation domain corrections, such as LC gradient or MS/MS transitions (de)-optimization, can be investigated to obtain a balanced response. If SIL is used as surrogate calibrant, the analyst should explore the potential presence of crosstalk interferences such as isotopic pattern overlap or impurities coming from SIL standards [8]. In MS, crosstalk occurs when ions from one scan event are still present in the collision cell when a second transition is taking place. This leads to signal artifacts in the next transition's chromatogram.

The *RF* can diverge from unit value when SILs containing enriched hydrogen atoms are used, but as long as the unit value slope remains within the $\pm 15\%$ acceptance interval compared to the authentic analyte slope, investigated SILs can be selected as surrogate calibrants. For example, tryptophan was successfully quantified in plasma with a relative bias between -2.0 and -8.0% using its deuterated analogue, even if the response factor was 0.67 [9]. Once the *RF* has been established, a multi-point calibration is performed in a pooled authentic matrix and the concentration of the authentic analyte is computed as follows:

Corrected concentration of authentic analyte

$$Z = \frac{\left(\frac{Y}{Y_c}\right) - a_0}{RF \times a_1} \quad (1.3)$$

where:

- Y and Y_c refer to the measured signal Y of the authentic analyte and the IS, respectively.
- Coefficients a_1 and a_0 characterize the slope and intercept of the calibration line performed with the surrogate standard.

Likewise, MMEC's use of an IS remains strongly recommended to correct for sample preparation and matrix effect variation between working samples and calibrators, thus improving trueness and precision when dealing with routine sample determination. Because the endogenous concentration of the authentic analyte in a pooled matrix is stable, an exciting possibility to implement this quantification method is to use this signal as an IS to normalize the instrument response of the surrogate standard calibration.

This approach, called Isotope Inversion, provides the same quantitative results for steroid determination as using the authentic analyte in a surrogate matrix such as active-charcoal stripped serum in this application [10]. When no signal from endogenous analyte interferes with the surrogate signal, the surrogate calibrant in authentic matrix can be a suitable alternative to the matrix-matched external calibration, especially when high endogenous concentration is present and/or intra- and inter-sample variations are observed.

1.3.4 Surrogate Calibrant in Surrogate Matrix

The increased commercial availability of SILs has raised interest in their use as surrogate calibrants in surrogate matrices to reduce calibration preparation time. Numerous publications have demonstrated their benefit, especially when MS detection is considered. This semi-targeted quantitation approach could be used to determine the amount of target analytes without needing authentic chemical standards. For instance, exogenous compounds were selected as potential surrogate calibrants in several biological matrices such as blood, plasma, urine, cerebrospinal fluid, and tissue homogenate [11]. In some cases, the combination of the surrogate calibrant in surrogate matrix allows extending the number of analytes that can be quantified in a single analysis.